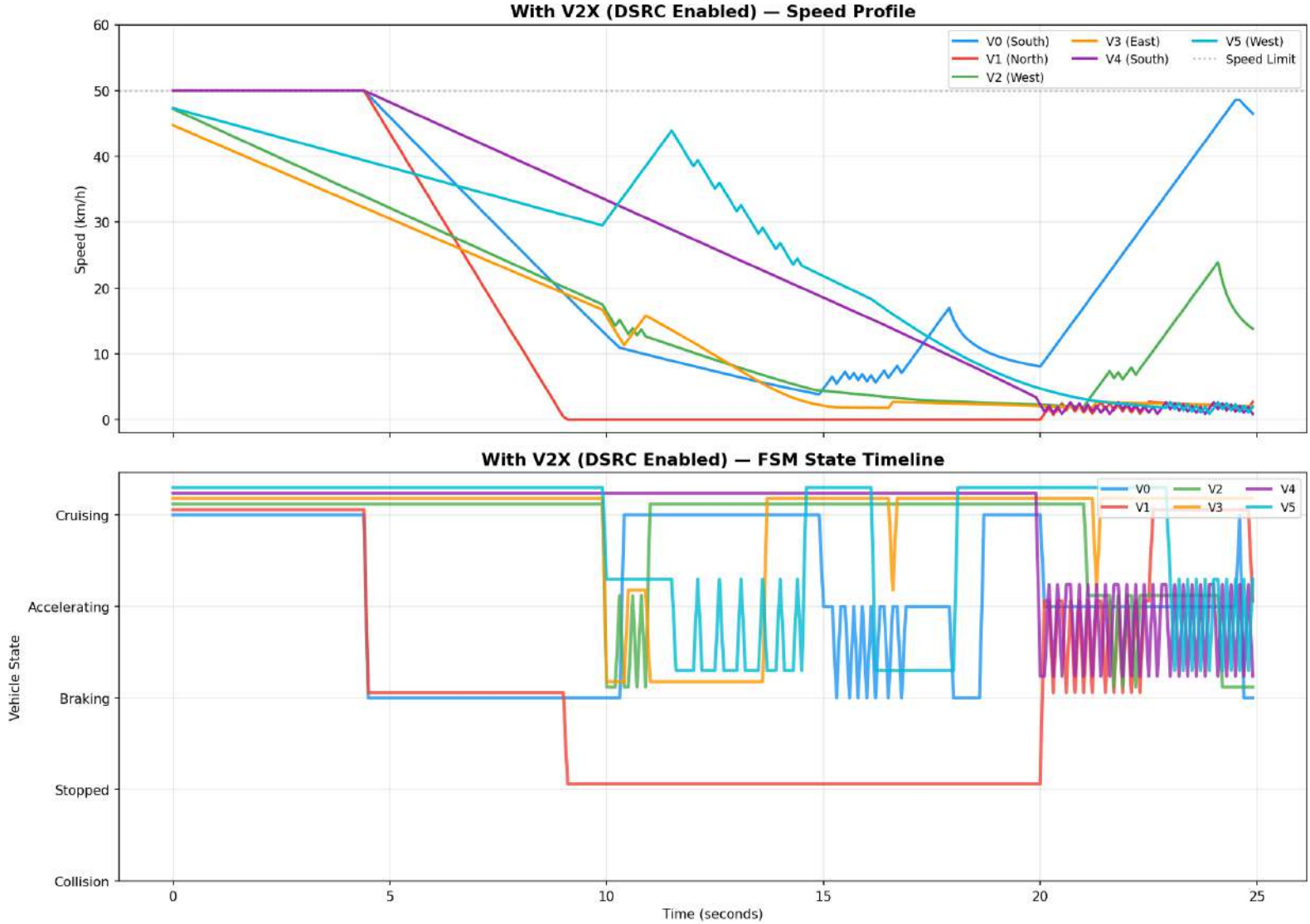
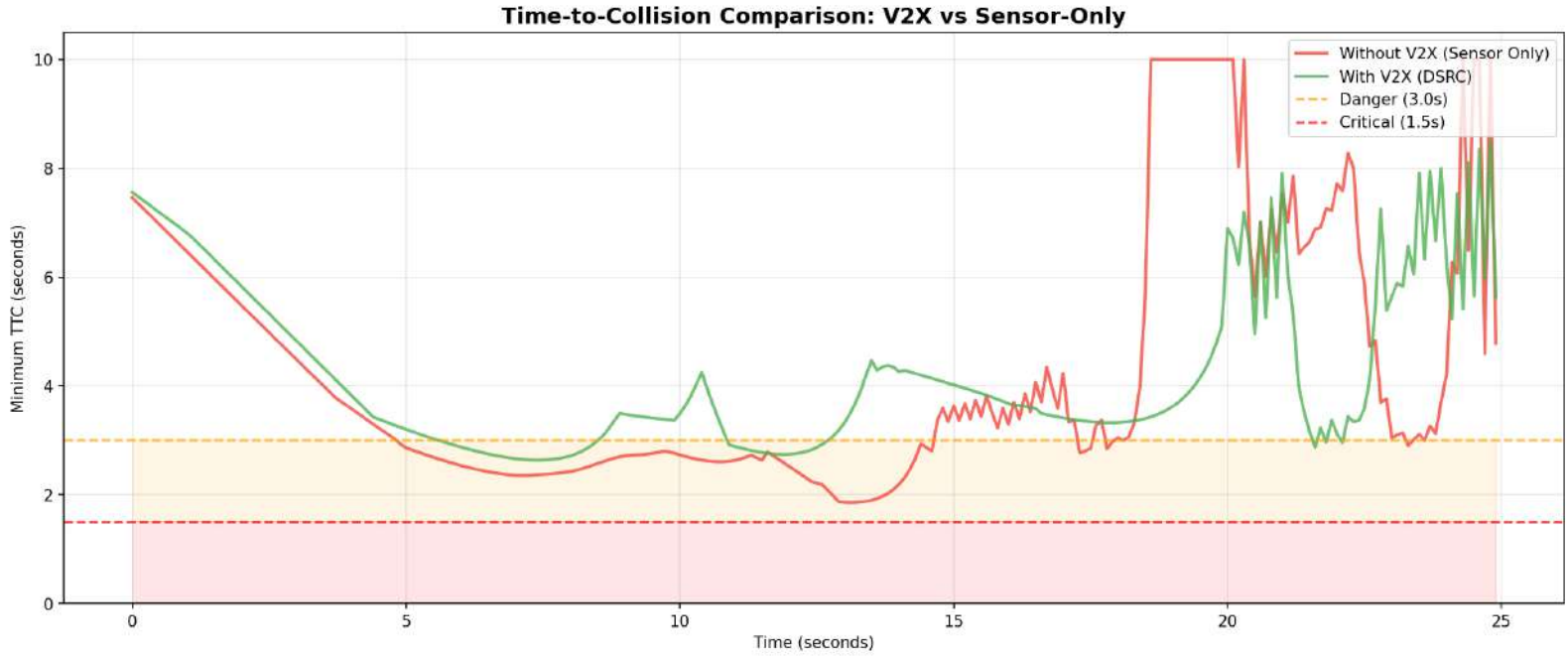
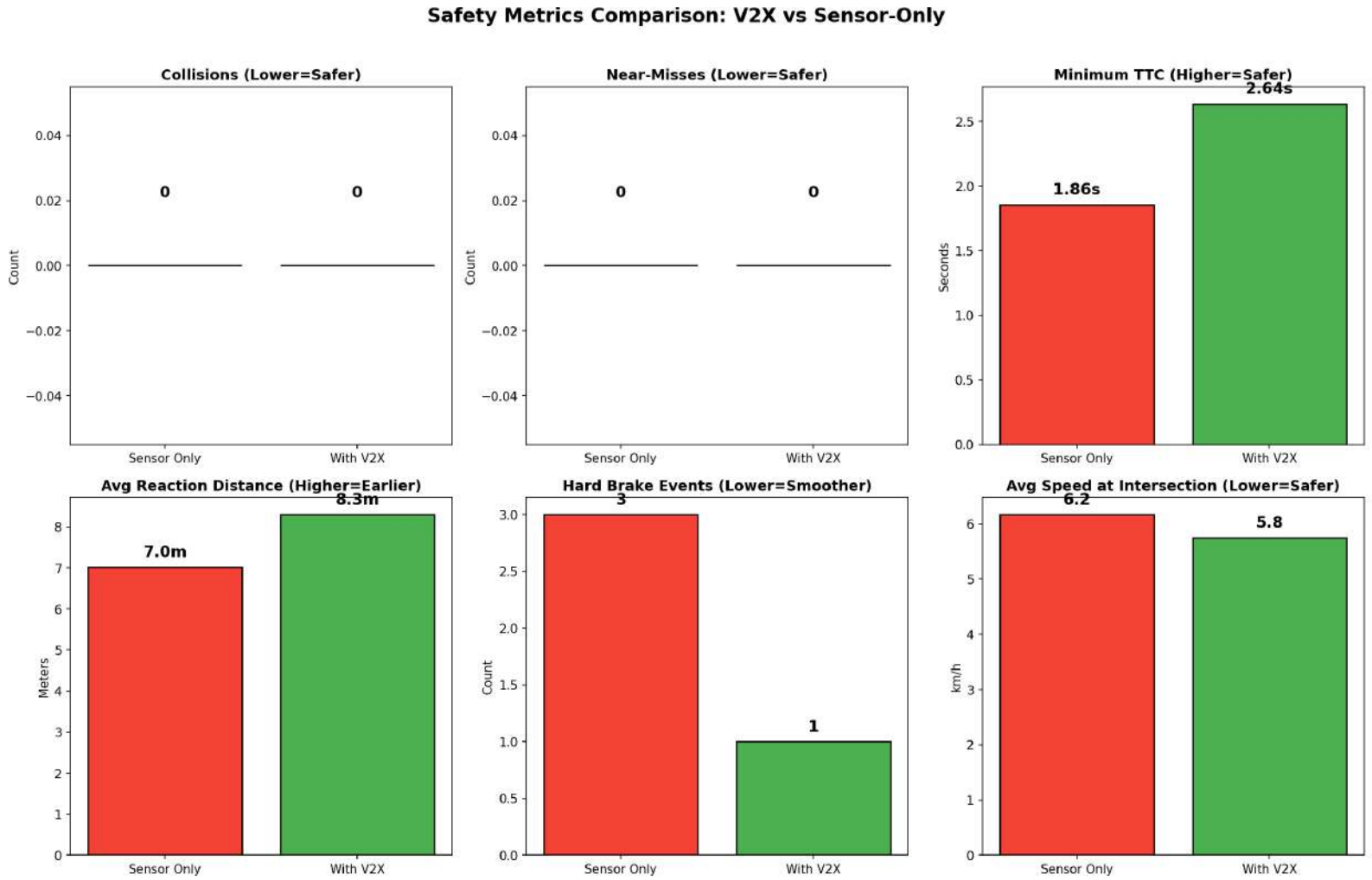
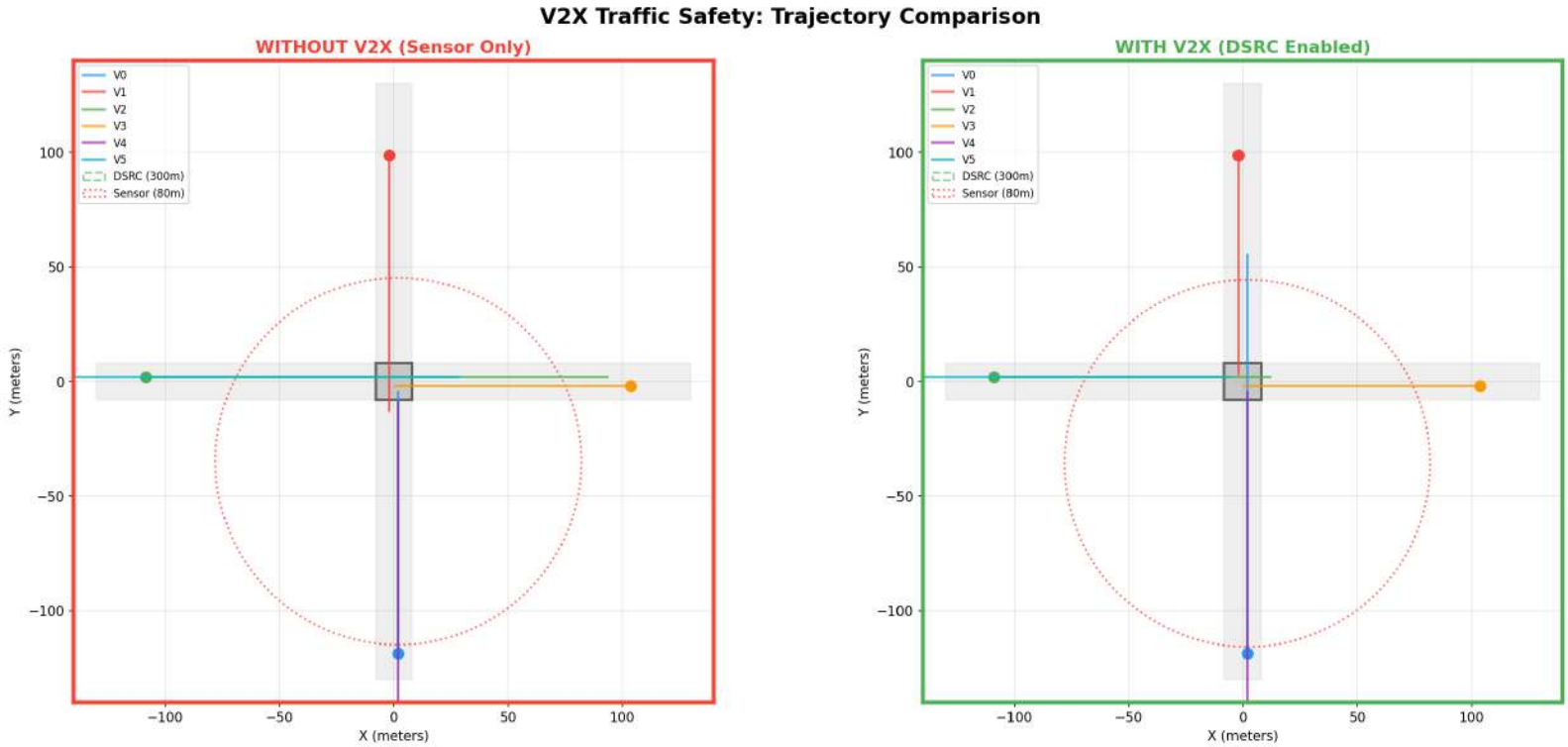
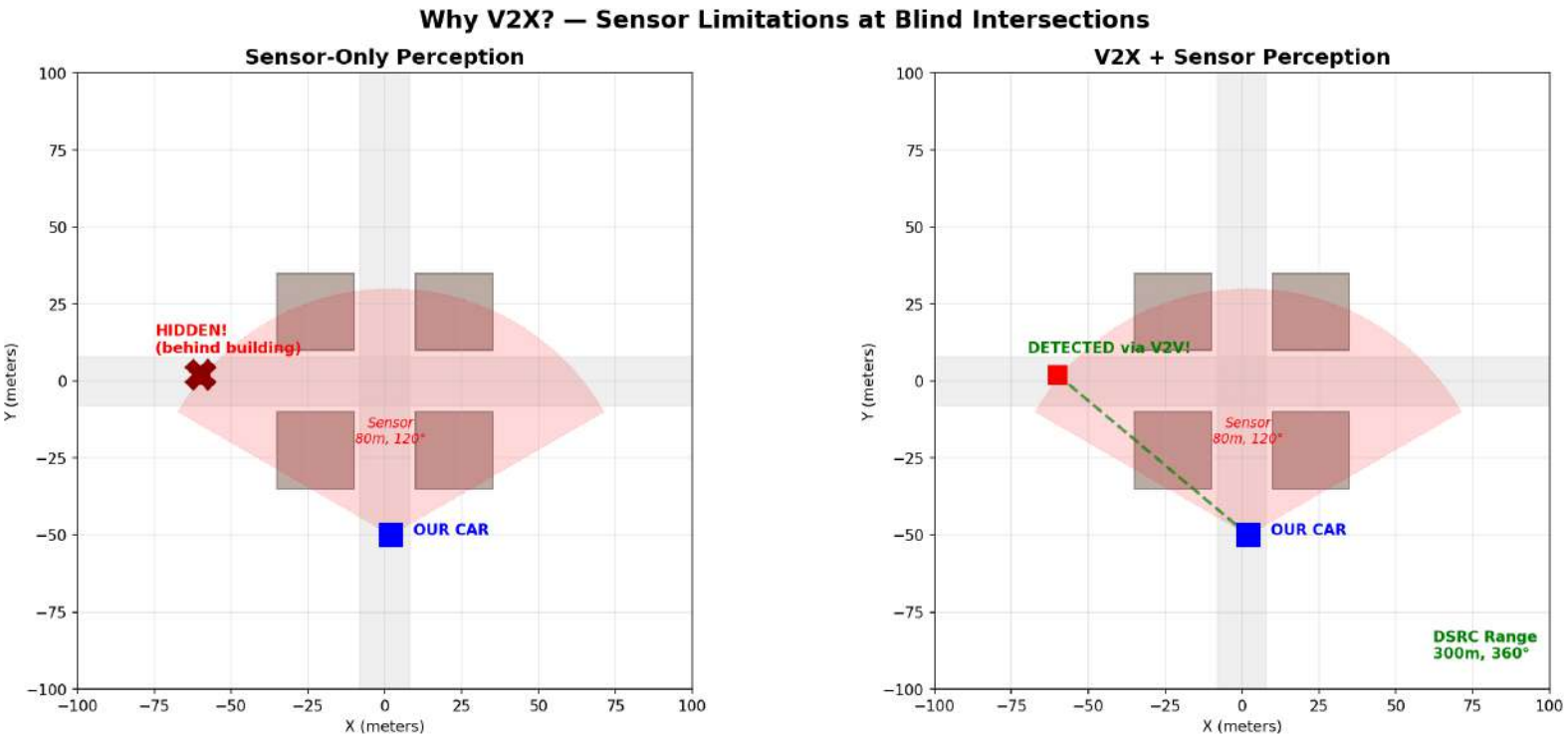
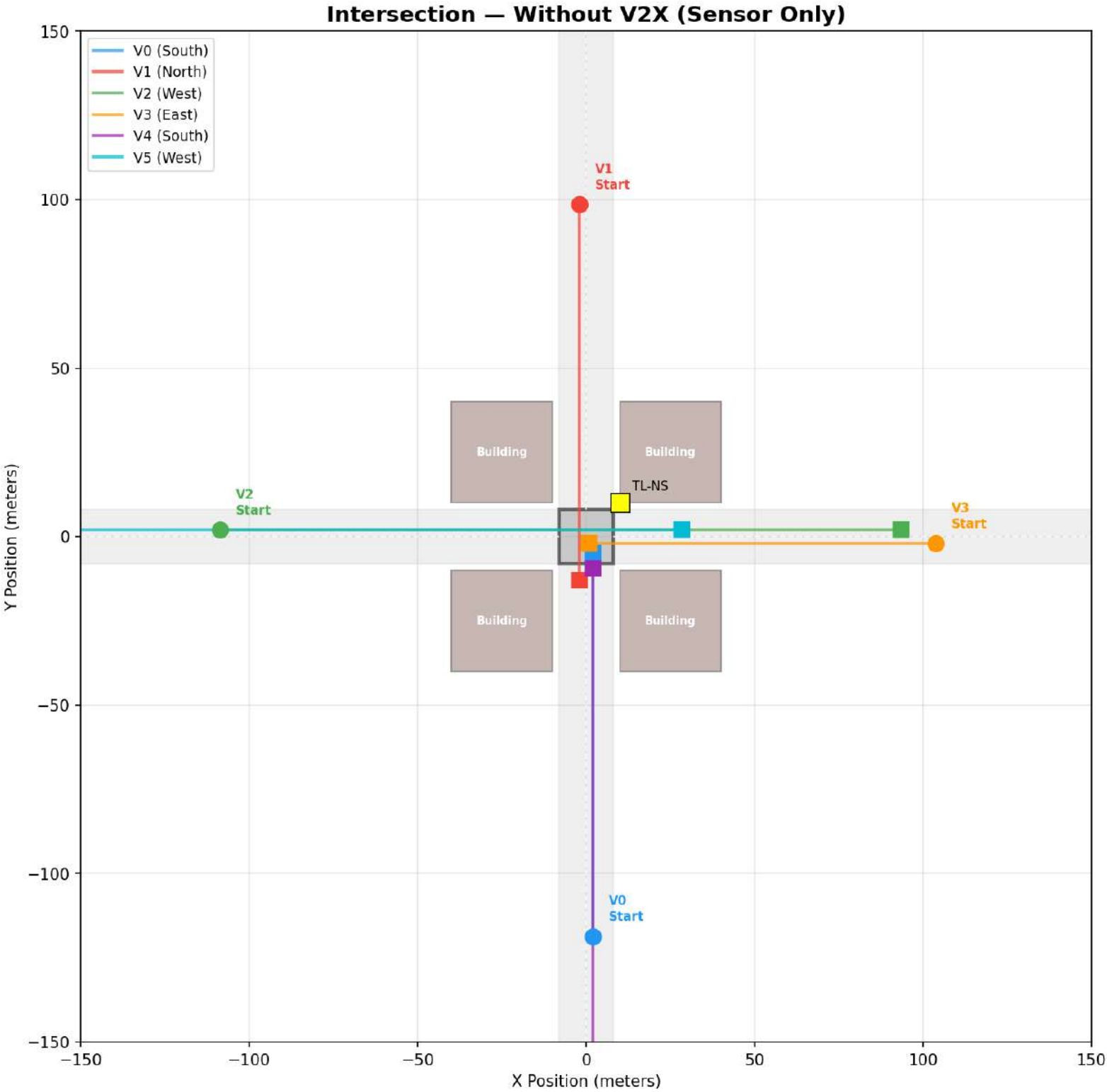

With V2X (DSRC Enabled) — Speed Profile
Speed (km/h)
V0 (South)
V1 (North)
V2 (West)
V3 (East)
V4 (South)
V5 (West)
Speed Limit
With V2X (DSRC Enabled) — FSM State Timeline
Vehicle State
Cruising
Accelerating
Braking
Stopped
Collision
Time (seconds)

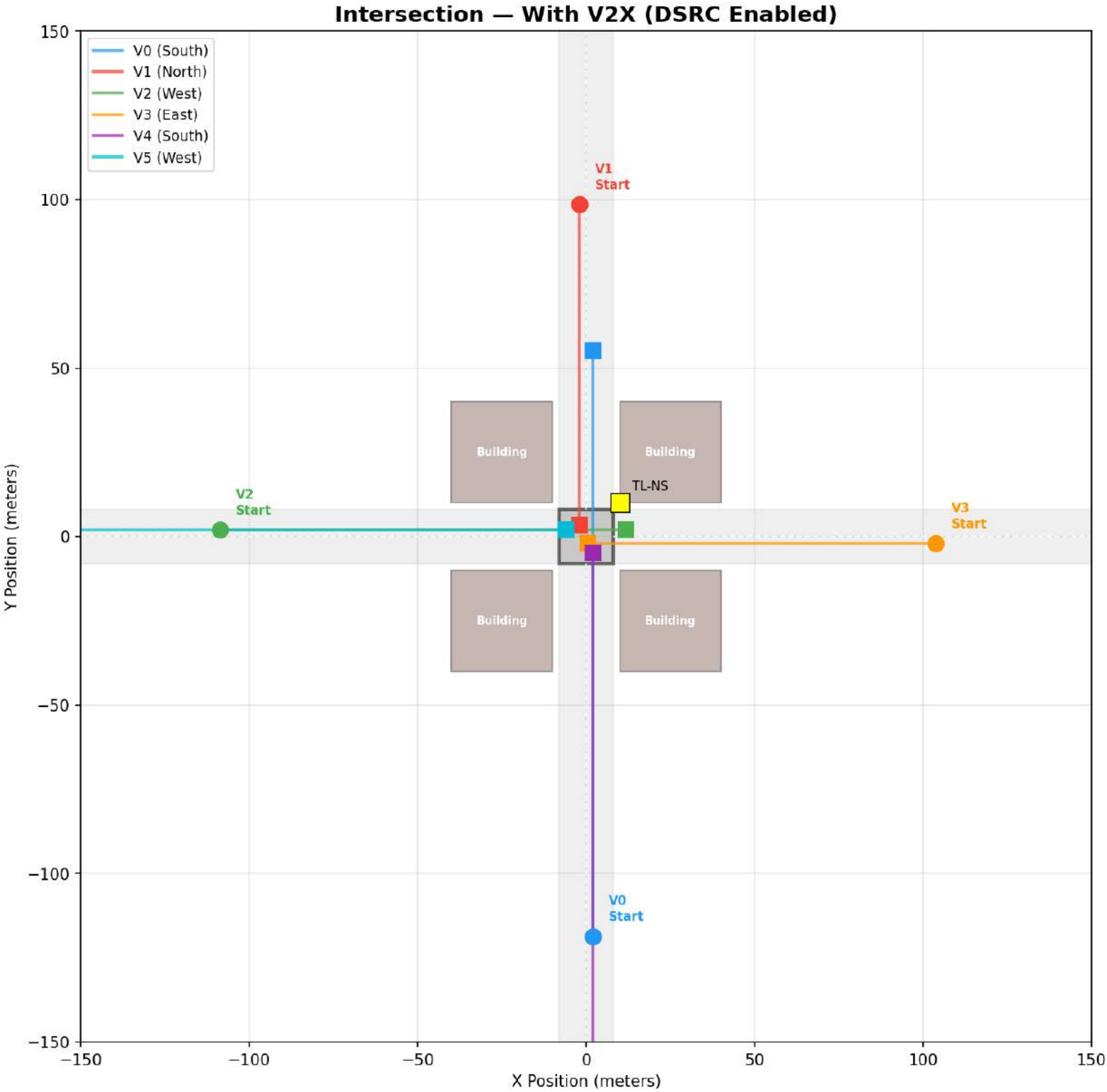

Time-to-Collision Comparison: V2X vs Sensor-Only
Without V2X (Sensor Only)
With V2X (DSRC)
Danger (3.0s)
Critical (1.5s)
Minimum TTC (seconds)
Time (seconds)
0
2
4
6
8
10
0
5
10
15
20
25

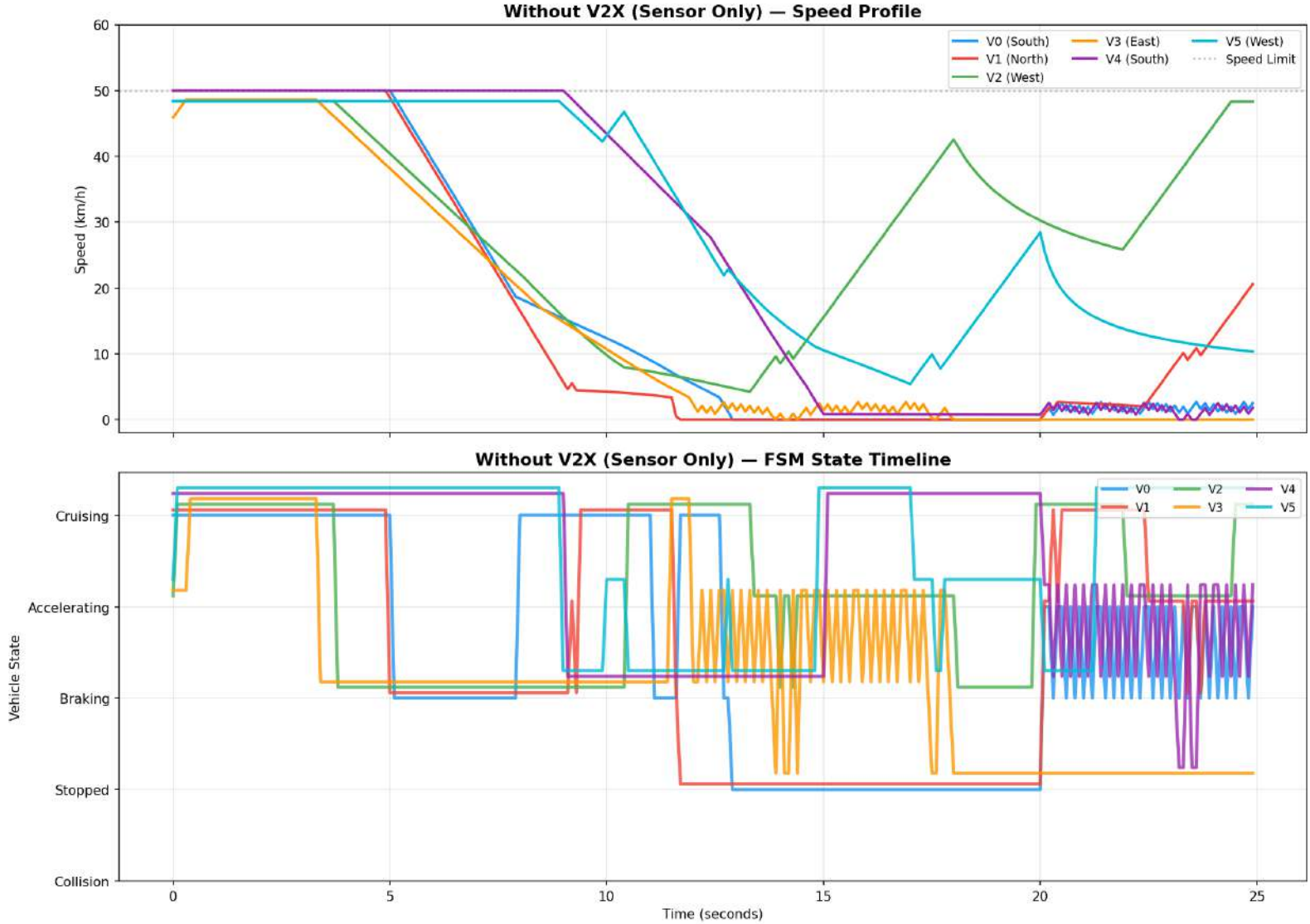

Safety Metrics Comparison: V2X vs Sensor-Only
Collisions (Lower=Safer)
Count
0
0
Sensor Only
With V2X
Near-Misses (Lower=Safer)
Count
0
0
Sensor Only
With V2X
Minimum TTC (Higher=Safer)
Seconds
1.86s
2.64s
Sensor Only
With V2X
Avg Reaction Distance (Higher=Earlier)
Meters
7.0m
8.3m
Sensor Only
With V2X
Hard Brake Events (Lower=Smoother)
Count
3
1
Sensor Only
With V2X
Avg Speed at Intersection (Lower=Safer)
km/h
6.2
5.8
Sensor Only
With V2X


V2X Traffic Safety: Trajectory Comparison
WITHOUT V2X (Sensor Only)
WITH V2X (DSRC Enabled)
V0
V1
V2
V3
V4
V5
DSRC (300m)
Sensor (80m)
Y (meters)
X (meters)
100
50
0
−50
−100


Why V2X? — Sensor Limitations at Blind Intersections
Sensor-Only Perception
HIDDEN!
(behind building)
Sensor
80m, 120°
OUR CAR
Y (meters)
X (meters)
V2X + Sensor Perception
DETECTED via V2V!
Sensor
80m, 120°
OUR CAR
DSRC Range
300m, 360°
Y (meters)
X (meters)


Intersection — Without V2X (Sensor Only)
V0 (South)
V1 (North)
V2 (West)
V3 (East)
V4 (South)
V5 (West)
V1
Start
V2
Start
V3
Start
V0
Start
TL-NS
Building
Building
Building
Building
Y Position (meters)
X Position (meters)
150
100
50
0
−50
−100
−150
−150
−100
−50
0
50
100
150


Intersection — With V2X (DSRC Enabled)
V0 (South)
V1 (North)
V2 (West)
V3 (East)
V4 (South)
V5 (West)
V1
Start
V2
Start
V3
Start
V0
Start
TL-NS
Building
Building
Building
Building
Y Position (meters)
X Position (meters)
150
100
50
0
−50
−100
−150
−150
−100
−50
0
50
100
150


Without V2X (Sensor Only) — Speed Profile
Speed (km/h)
V0 (South)
V1 (North)
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Speed Limit
Without V2X (Sensor Only) — FSM State Timeline
Vehicle State
Cruising
Accelerating
Braking
Stopped
Collision
Time (seconds)